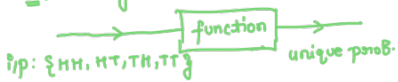
Random Variable

Random variable is a mathematical function.



i.e tossing two coin :



Input	HH	HT	TH	TT
Random Variable (X)	2	1	1	0

X : nos of HEADS/TAILS
↓
changing variable



$P[X=0H] = \frac{1}{4}$; $P[X=1H] = \frac{2}{4}$; $P[X=2H] = \frac{1}{4}$

A random variable is a variable that assigns a numerical value to the outcome of a random experiment.

i.e ② Tossing three coin

H H H	0T
H H T	1T
H T H	1T
H T T	2T
T H H	1T
T H T	2T
T T H	2T
T T T	3T

$x = 0, 1, 2, 3$

$P(x=r)$ random variable

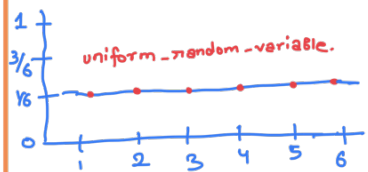
$P(x=0T) = 1/8$; $P(x=1T) = 3/8$

$P(x=2T) = 3/8$; $P(x=3T) = 1/8$

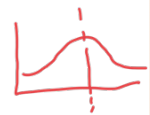
$\Sigma all = 1$

- Rolling a Balanced die :

X	1	2	3	4	5	6
P(x=n)	1/6	1/6	1/6	1/6	1/6	1/6

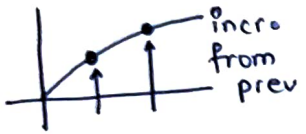


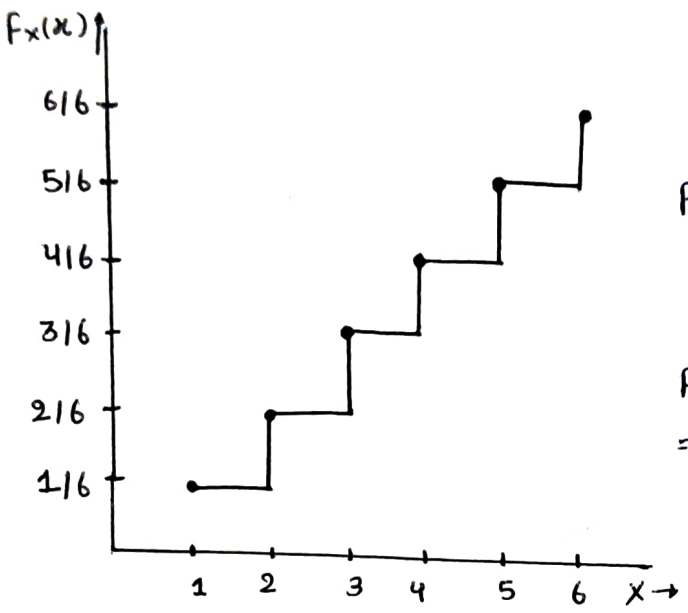
for large sample →



gaussian distribution

* Step-case / Ladder fxn [monotonically ↑↑]





* always +ve
  ↓
  $F(x_i) \geq 0$
  $P_i \geq 0$

* $F_X(\infty) = 0$
  $F_X(-\infty) = 0$

$F_X(x_i) = P[X \leq x_i]$

$F_X(1) = P[X \leq 1]$

$F_X(2) = P[X \leq 2]$
$= P[X=0] + P[X=1]$

⋮

$F_X(x_i) = P[X \leq x_i]$
$= P_1 + P_2 + P_3 + \cdots + P_i$

ie, rolling a die

X	1	2	3	4	5	6
$F_X(x_i)$	$F_X(1)$	$F_X(2)$	$F_X(3)$	$F_X(4)$	$F_X(5)$	$F_X(6)$
	1/6	2/6	3/6	4/6	5/6	6/6

Cumulative Distribution Function [CDF].

- The CDF of a discrete r.v. X is the probability that X takes a value less than or equal to a given value.

defined as: $F(x) = P(X \leq x)$

Outcome → The PMF give prob. at an individual point. while The CDF gives prob. upto a certain point.

probability mass function [PMF]

- The PMF of a random variable X gives the probability that X takes a specific value. Is defined as:

$P(X=x) = p(x)$ $p(x)$ is the prob. that the random variable X takes the value of x.

- The sum of all probabilities across all possible values of X must equal 1

$$\sum_{i=1}^{N} P[X=x_i] = 1$$

rolling a die,

$P(X=1) = 1/6$, $P(X=2) = 1/6$ ..... $P(X=6) = 1/6$

- finite or countable values ie, series of coin flip
  ↓
  (0H, 1H, 2H, ...)

Discrete random variable

Random variable

o can take upto infinite number (i.e, real nos b/w two endpoints).

continuous random variable

o since, a continuous rov can take an infinite nos of values, the prob. of variable taking an exact value is 0. Instead, it represent prob. density over an interval.

probability density function [pdf]

f(x)
pdf
(validity)

- $f(x) \geq 0 \quad \forall x$ [It's non-negative]
- Total Area under the curve of f(x) must equal to 1

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

o To find the prob. that rov lies within a specific interval :

$$P(a \leq X \leq B) = \int_{a}^{B} f(x)\,dx$$

$$F_x(B) - F_x(a) = \int_{a}^{B} f(x)\,dx$$

The cdf for a crv X is defined similarly to that of a drv. It, gives the prob. that the variable X is less than or equal to a particular value x.

The CDF, $F_x(x)$ is the integral of PDF from $-\infty$ to x

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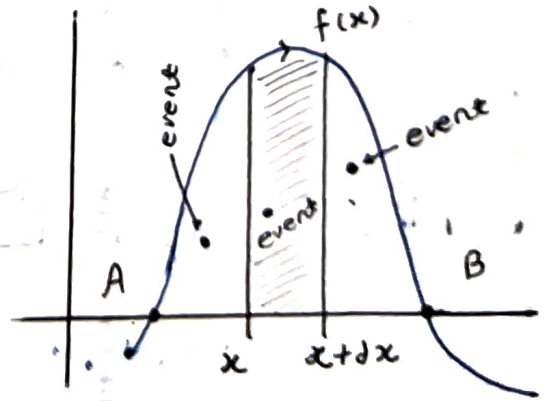
Continuous probability distribution

cdf for continuous variable.

$$F'_x(x) = f(x)$$

"or"

$$\frac{dF_x(x)}{dx} = f(x)$$



* Step-case / Ladder fxn

[monotonically ↑↑]



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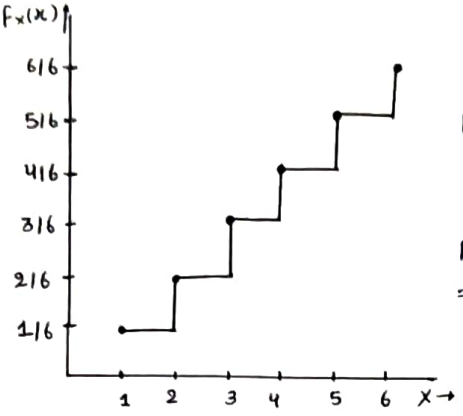
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$$F_X(x) = cdf = \int_R f(x)\,dx$$

↓

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